

CHAPTER 6

MOMENTS

1.5.5 Turning effect of forces

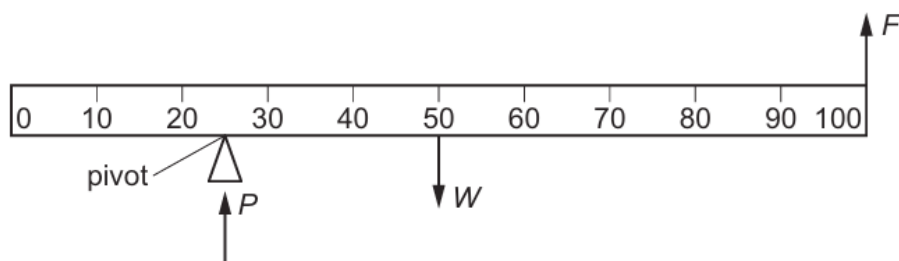
- 1 Describe the moment of a force as a measure of its turning effect and give everyday examples
- 2 Define the moment of a force as $\text{moment} = \text{force} \times \text{perpendicular distance from the pivot}$; recall and use this equation
- 3 State and use the principle of moments for an object in equilibrium
- 4 Describe an experiment to verify the principle of moments

1.5.6 Centre of gravity

- 1 State what is meant by centre of gravity
- 2 Describe how to determine the position of the centre of gravity of a plane lamina using a plumb line
- 3 Describe, qualitatively, the effect of the position of the centre of gravity on the stability of simple objects

1. O/N/23/11/Q7

A uniform metre rule of weight W is pivoted at the 25 cm mark and held horizontal by a force F applied upwards at the 100 cm mark. The rule is supported by a vertical force P acting at the pivot.

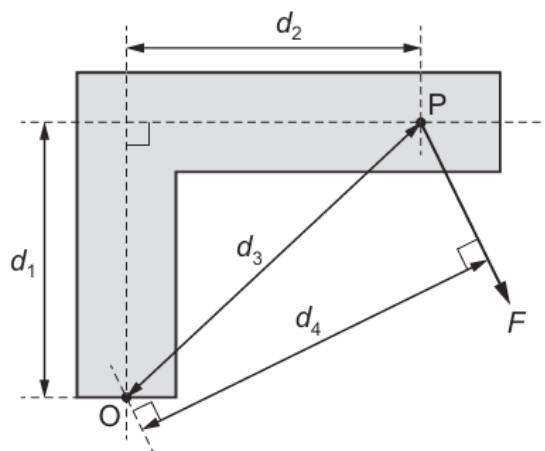


What is the magnitude of P ?

- A** $\frac{1}{3} W$ **B** $\frac{1}{2} W$ **C** $\frac{2}{3} W$ **D** $\frac{4}{3} W$

2. O/N/23/11/Q6

A force F causes a moment about point O on an L-shaped bar. The force F acts at point P.

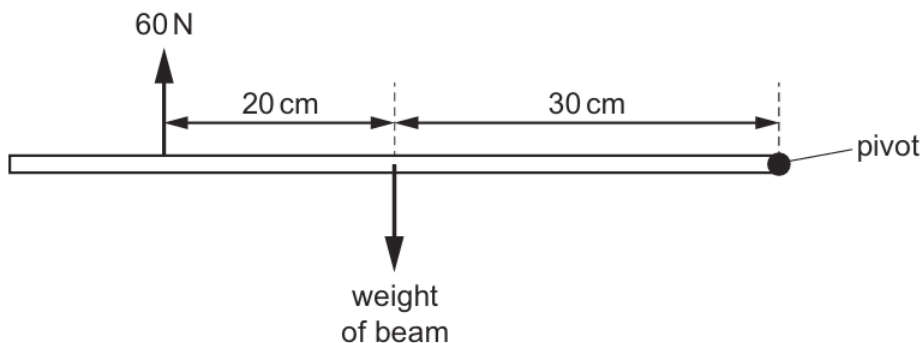


What is the moment of F about O?

- A** Fd_1 **B** Fd_2 **C** Fd_3 **D** Fd_4

3. M/J/23/11/Q9

A uniform horizontal beam, pivoted at its right-hand end, is in equilibrium. A force of 60 N acts vertically upwards on the beam as shown.

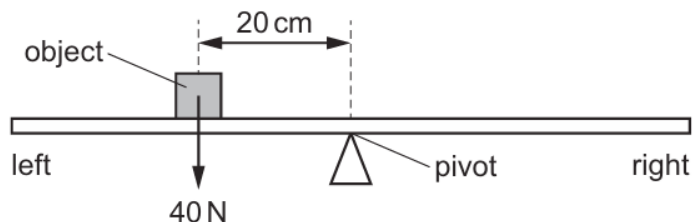


What is the weight of the beam?

- A** 36 N **B** 40 N **C** 90 N **D** 100 N

4. M/J/23/12/Q11

A uniform beam is pivoted at its midpoint. An object weighing 40 N is placed on the beam 20 cm to the left of the pivot as shown.

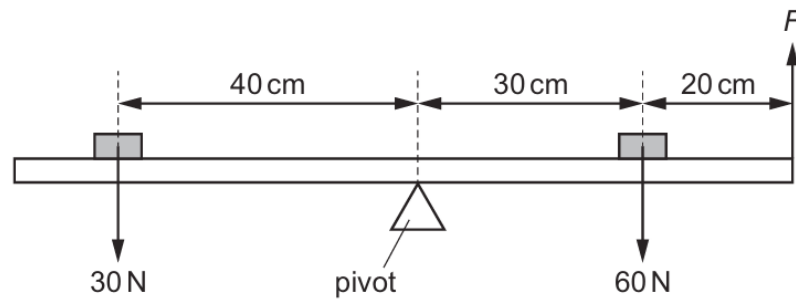


Which force and position balances the system?

- A** 20 N acting downwards, 40 cm to the right of the pivot
B 20 N acting upwards, 40 cm to the right of the pivot
C 50 N acting downwards, 10 cm to the left of the pivot
D 50 N acting upwards, 10 cm to the left of the pivot

5. O/N/22/11/Q11

A uniform beam is pivoted at its centre. Two weights are placed on the beam in the positions shown and the beam is balanced by an upward force F .



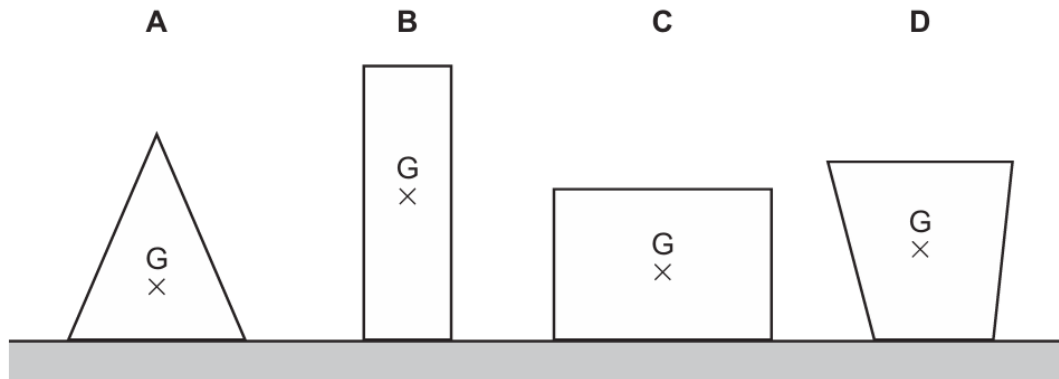
What is the size of F ?

- A** 6.0 N **B** 12 N **C** 30 N **D** 60 N

6. O/N/22/12/Q13

Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

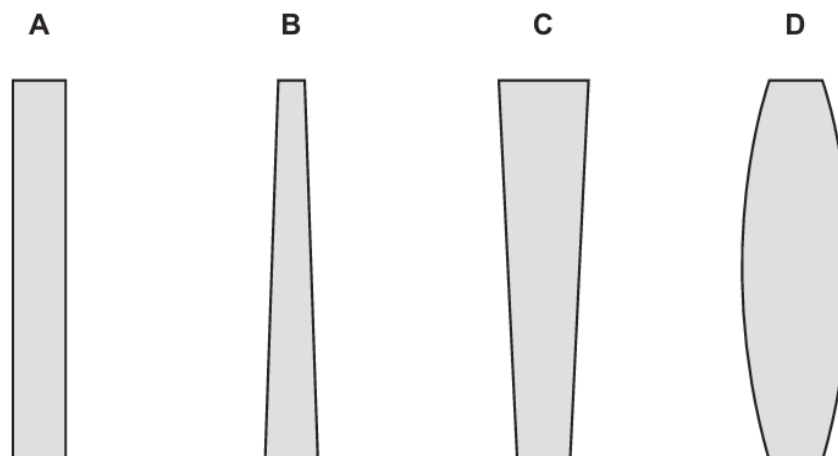
Which object is the least stable?



7. M/J/22/11/Q6

Four glass objects have square bases of equal area.

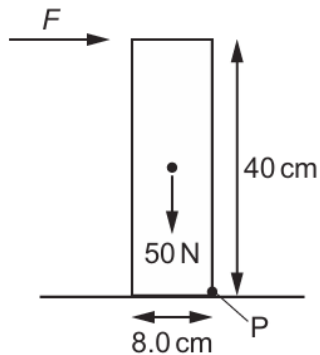
Which object is the least stable?



8. M/J/22/12/Q7

The diagram shows a uniform solid rectangular block of weight 50 N that is pivoted about point P.

The height of the block is 40 cm. The base of the block is 8.0 cm wide.

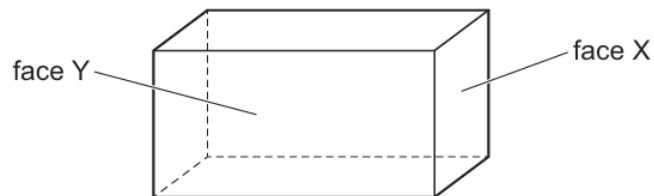


Which horizontal force F just makes the block start to rotate about P?

- A** 2.5 N **B** 5.0 N **C** 10 N **D** 160 N

9. M/J/22/12/Q8

The centre of mass of a solid rectangular block is at its centre. A small heavy weight is available.



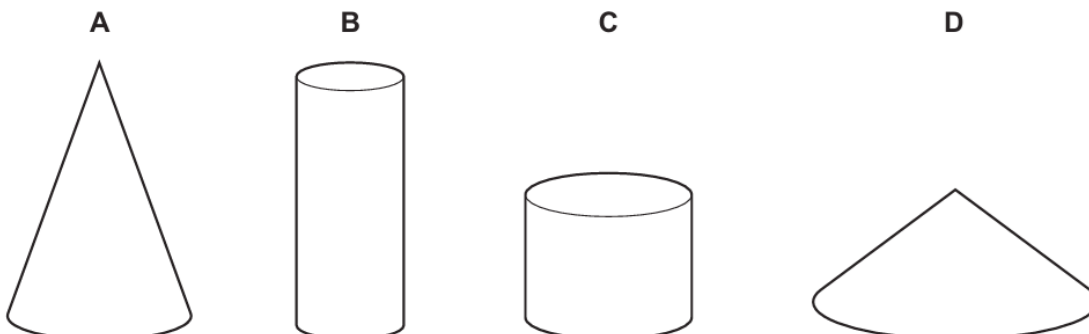
In which arrangement is the centre of mass the lowest?

- A** with face X on a table
B with face Y on a table
C with face X on a table and the heavy weight attached centrally on top of the block
D with face Y on a table and the heavy weight attached centrally on top of the block

10. O/N/21/12/Q8

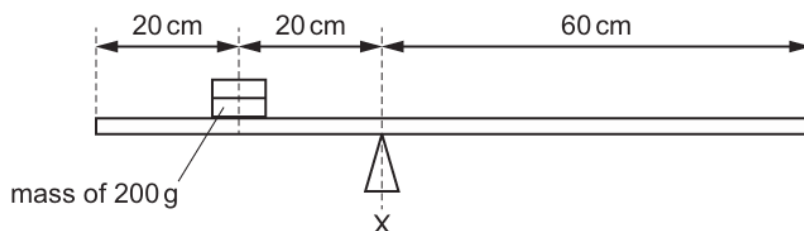
Four solid objects are made from the same material and have equal volumes.

Which object is the most stable?



11. M/J/21/11/Q8

A horizontal beam is pivoted at X. A mass of 200 g rests on the beam as shown. The centre of mass of the beam is 50 cm from the right-hand end of the beam.



The beam is balanced.

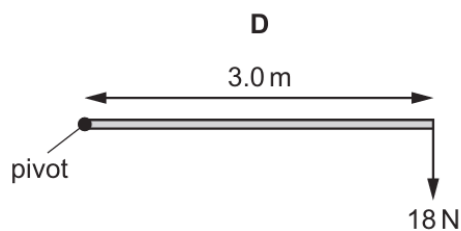
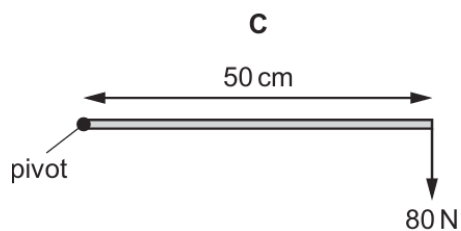
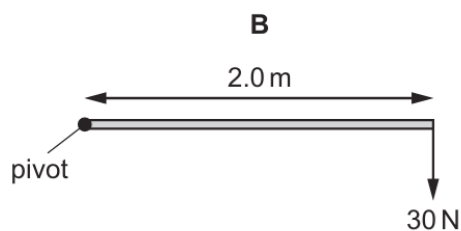
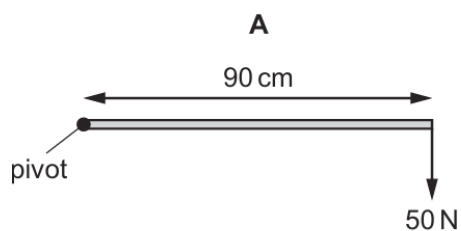
What is the mass of the beam?

- A** 80 g **B** 100 g **C** 400 g **D** 800 g

12. M/J/21/12/Q8

Some forces are applied at different distances from a pivot.

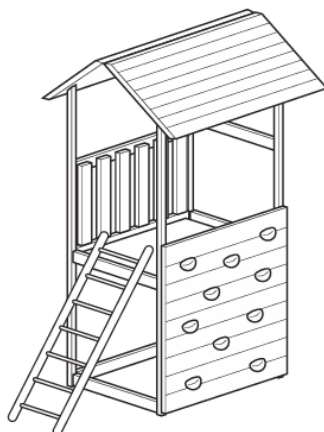
Which diagram shows the force that produces the largest moment about the pivot?



not to scale

13. M/J/21/12/Q9

The diagram shows a children's wooden play tower.

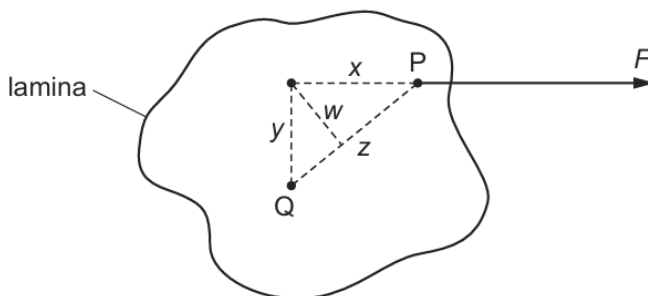


Which change to the tower makes it more stable?

- A** making the tower narrower
- B** making the tower taller
- C** lowering the centre of mass
- D** raising the centre of mass

14. O/N/20/11/Q10

A length of thread is attached to a lamina at point P, as shown in the diagram.



The lamina is free to rotate about point Q.

The tension in the thread is F .

What is the moment of F about Q?

- A** Fw
- B** Fx
- C** Fy
- D** Fz

15. M/J/20/12/Q11

Which single item can be used to find the centre of mass of a plane lamina of irregular shape?

- A** a balance
- B** a measuring tape
- C** a micrometer
- D** a vertical pin

16. O/N/19/11/Q11

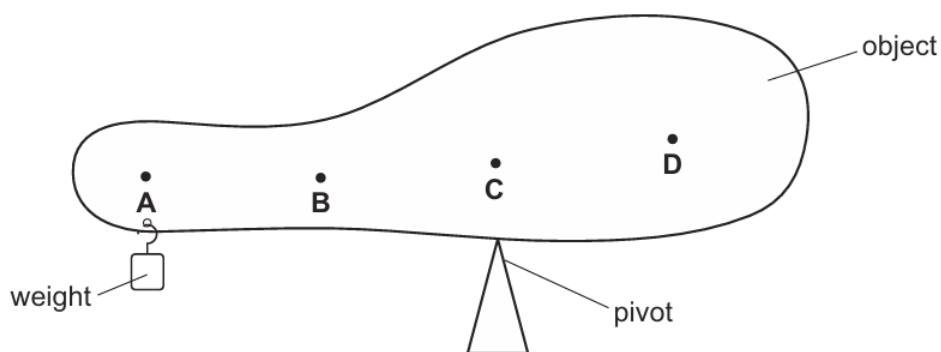
What is an expression for the moment of a force about a pivot?

- A distance the force moves from the pivot $\div$ force
- B force $\times$ distance the force moves from the pivot
- C force $\times$ perpendicular distance of the force from the pivot
- D force $\div$ perpendicular distance of the force from the pivot

17. O/N/19/11/Q12

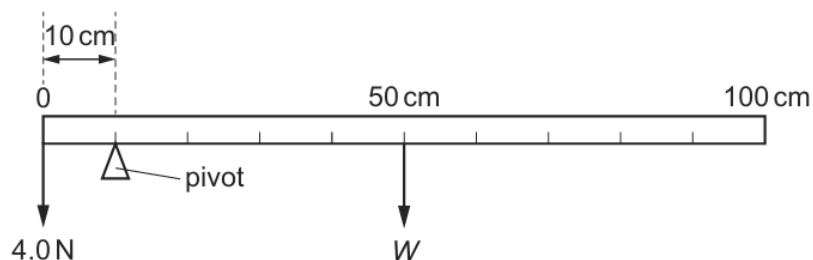
A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object, as shown.

Where is the centre of mass of the object?



18. M/J/19/11/Q9

A uniform metre rule is balanced by a 4.0 N weight as shown.



What is the weight W of the metre rule?

- A 1.0 N
- B 4.0 N
- C 16 N
- D 40 N

19. M/J/19/11/Q10

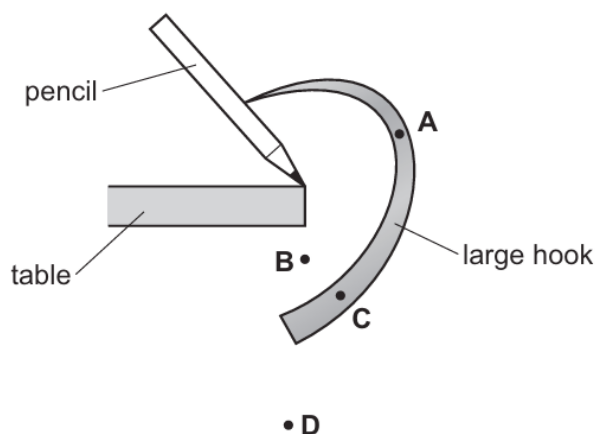
Which statement about centre of mass is correct?

- A Objects with a centre of mass at the same height are less stable when the base is larger.
- B Objects with a centre of mass at the same height are more stable when the base is larger.
- C Objects with higher centres of mass and smaller bases are more stable.
- D Objects with identical bases are more stable when the centre of mass is higher.

20. M/J/19/12/Q11

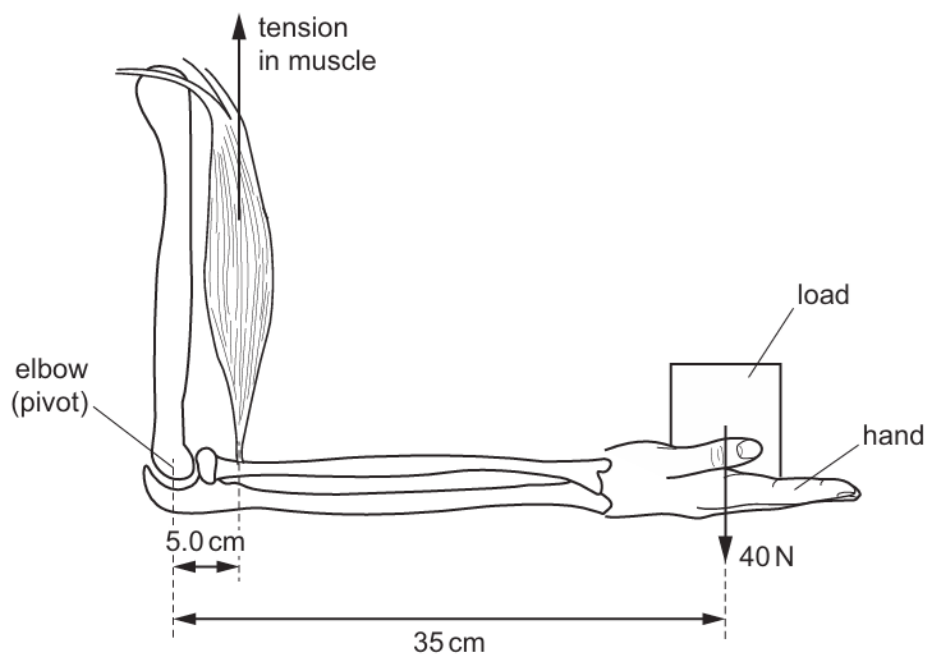
A large hook sticks into a pencil. The hook and pencil are balanced on the edge of a table, as shown.

Where is the centre of mass of the hook and pencil?



21. M/J/18/11/Q9

The diagram shows a muscle and bones in a person's arm. The hand holds a load of weight 40 N. The elbow acts as a pivot and the tension in the muscle keeps the lower part of the arm horizontal.



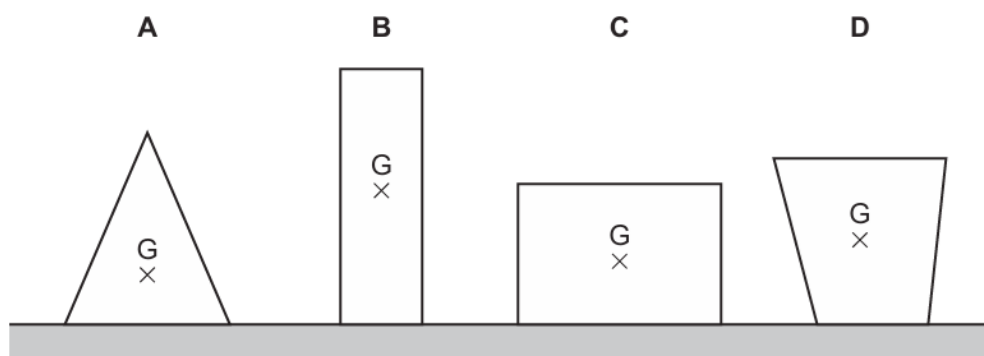
What is the tension in the muscle due to the load?

- A** 200 N **B** 240 N **C** 280 N **D** 1400 N

22. M/J/18/11/Q10

Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

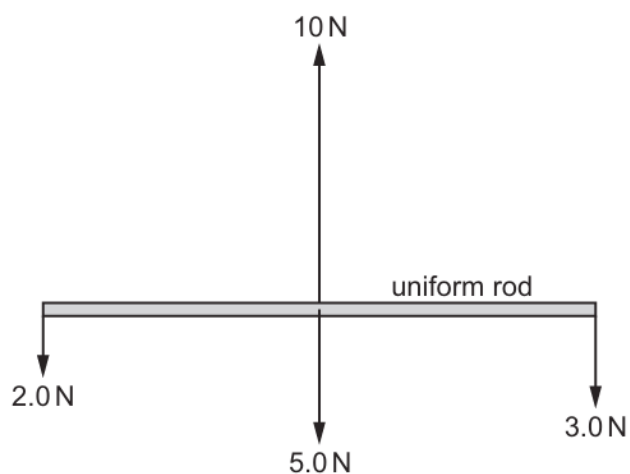
Which object is the least stable?



23. M/J/18/12/Q7

A uniform rod of weight 5.0 N is held initially at rest.

The diagram shows the forces acting on the rod when it is released.



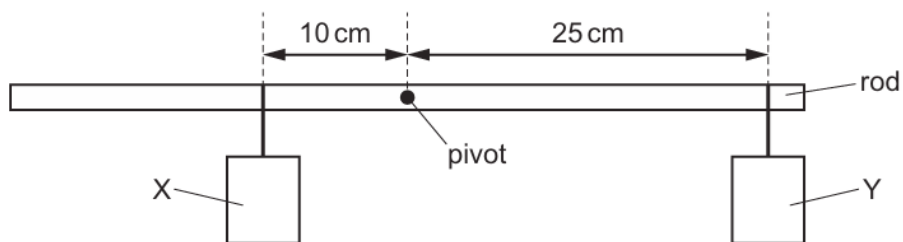
What happens to the rod when it is released?

- A** It does not move.
- B** It moves to the right.
- C** It moves upwards.
- D** It starts to rotate.

24. M/J/18/12/Q10

Two objects X and Y are suspended from a uniform rod, pivoted at its centre.

The rod is in equilibrium.

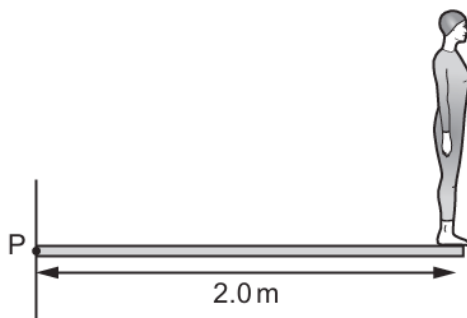


Which statement about X and Y is correct?

- A** The mass of X is 0.4 times the mass of Y.
- B** The mass of X is 2.5 times the mass of Y.
- C** The mass of X is 3.5 times the mass of Y.
- D** The mass of X is equal to the mass of Y.

25. O/N/17/12/Q7

A diver of weight 500 N stands at the end of a springboard that is 2.0 m long and is fixed at point P.



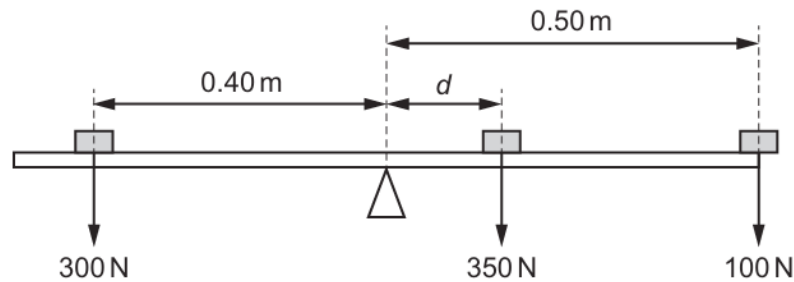
The springboard has a weight of 500 N. The centre of mass of the springboard is in the centre of the board.

What is the total moment about point P of the diver and the board?

- A** 500 N m **B** 750 N m **C** 1000 N m **D** 1500 N m

26. M/J/17/11/Q12

A uniform beam is pivoted at its centre. The beam is balanced by three weights in the positions shown.

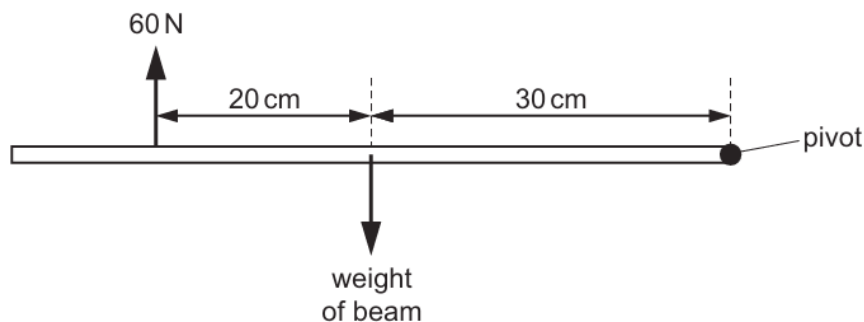


What is the length d ?

- A** 0.020 m **B** 0.050 m **C** 0.20 m **D** 0.48 m

27. M/J/16/11/Q9

A uniform horizontal beam, pivoted at its right-hand end, is in equilibrium. A force of 60 N acts vertically upwards on the beam as shown.



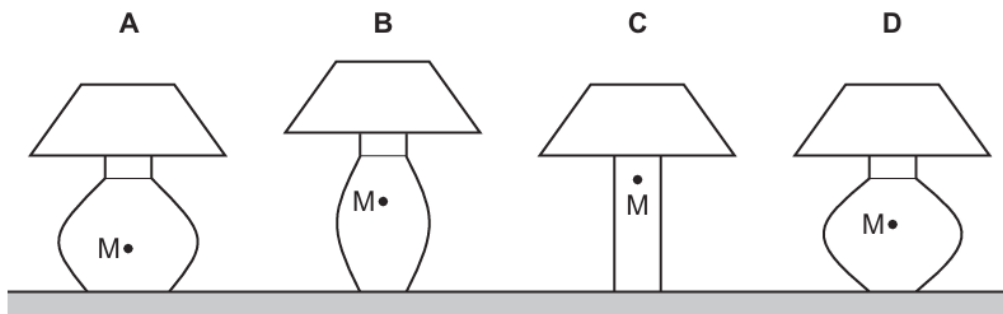
What is the weight of the beam?

- A** 36 N **B** 40 N **C** 90 N **D** 100 N

28. M/J/16/11/Q10

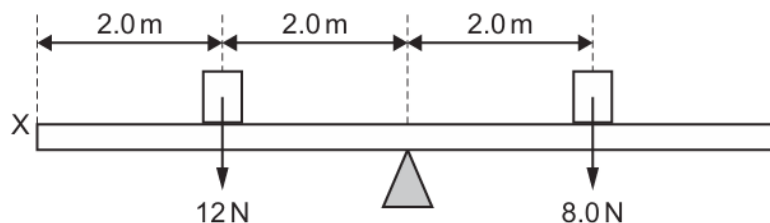
Four table lamps are shown along with the position M of the centre of mass in each case.

Which lamp is the most stable?



29. M/J/16/12/Q6

A uniform plank is pivoted at its mid-point. Two weights are added to the plank, one weight on each side of the pivot in the positions shown.



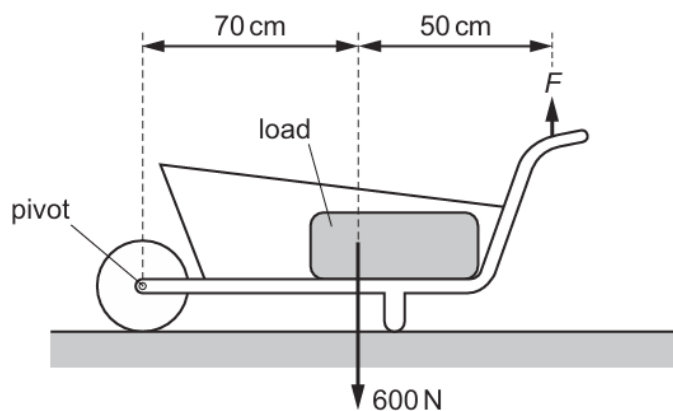
A vertical force is applied at point X to balance the plank.

What is the size and direction of this force?

	size / N	direction
A	2.0	downwards
B	2.0	upwards
C	4.0	downwards
D	4.0	upwards

30. O/N/15/11/Q8

The total weight of the load and the wheelbarrow shown is 600 N.



What is the size of force F needed just to lift the loaded wheelbarrow?

- A** 350 N **B** 430 N **C** 600 N **D** 840 N

31. M/J/15/11/Q10

When a car turns a corner at speed, it risks toppling over. Two factors affecting the stability of a car are the height of its centre of mass and the distance between its front wheels.

Which factors make the car most stable?

	centre of mass	distance between front wheels
A	high	small
B	high	large
C	low	small
D	low	large

32. O/N/14/11/Q7

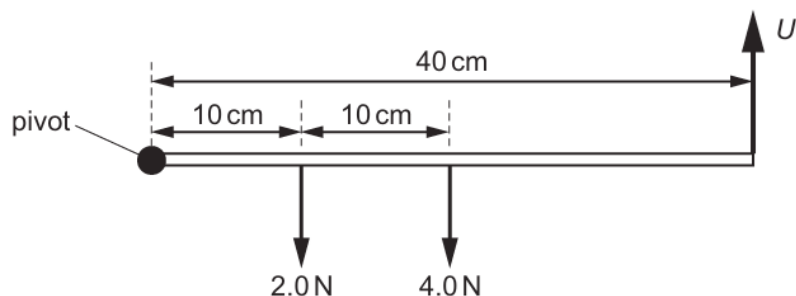
What affects the stability of an object?

- A** only its base area and the location of its centre of mass
- B** only its weight and its base area
- C** only the location of its centre of mass
- D** only its weight

33. M/J/14/11/Q9

A beam of length 40 cm is pivoted at one end.

The weight of the beam is 4.0 N and acts at a point 20 cm from the pivot. A 2.0 N weight hangs 10 cm from the pivot.



An upward force U is needed to keep the beam horizontal.

What is the size of U ?

- A** 0.5 N
- B** 1.5 N
- C** 2.5 N
- D** 6.0 N

34. M/J/14/11/Q10

A car is designed to be stable.

To achieve good stability, where is the centre of mass of the car?

- A** above the front wheels
- B** above the rear wheels
- C** as high in the car as possible
- D** as low in the car as possible

35. M/J/14/12/Q10

A man uses clay to make a pot. He wants the pot to be as stable as possible when placed on a flat surface.

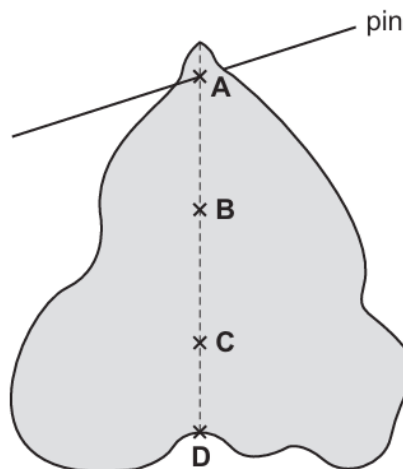
Which two features of the pot must the man consider?

- A** the area of the base and the height of the centre of gravity
- B** the density of the clay and the area of the base
- C** the density of the clay and the height of the centre of gravity
- D** the weight and the height of the centre of gravity

36. O/N/13/11/Q7

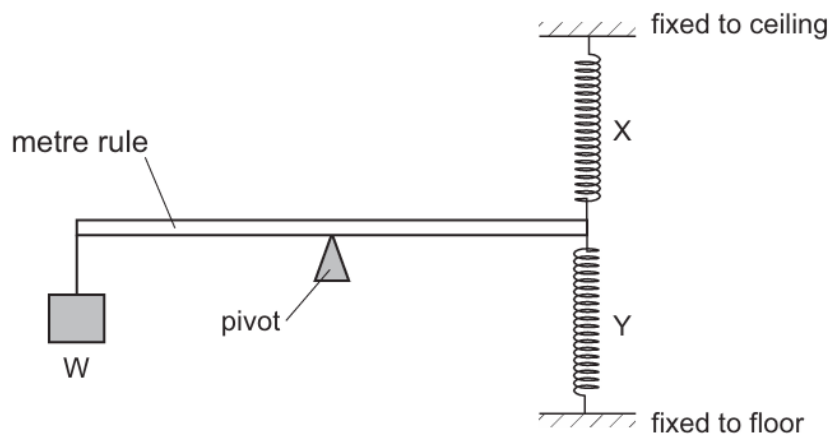
A piece of uniform card is suspended freely from a horizontal pin.

Which point is its centre of mass?



37. O/N/13/12/Q7

Two stretched springs X and Y are attached to one end of a metre rule as shown. A weight W is hung from the other end. A pivot is at the centre of the rule.



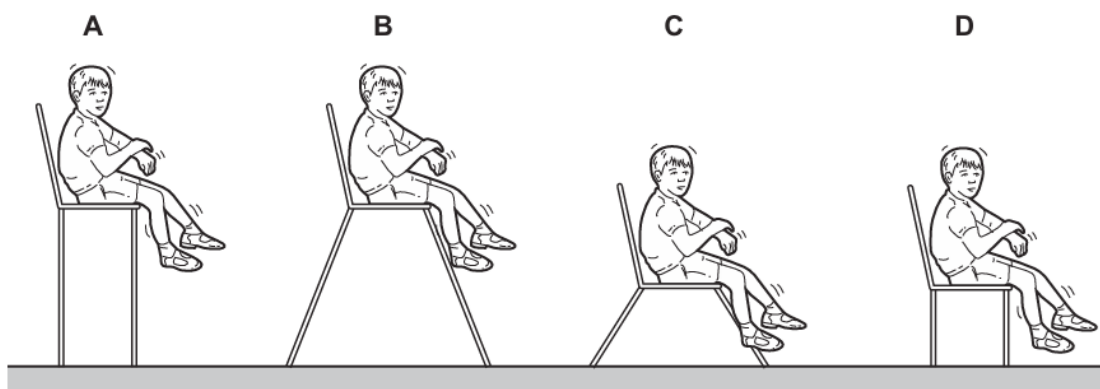
The weight W is moved towards the pivot.

How does the extension of each spring change?

	spring X	spring Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

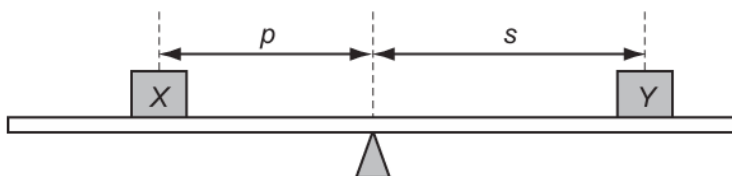
38. M/J/13/11/Q7

Which chair is the **least** stable if the child moves?



39. O/N/12/11/Q7

Masses X and Y are placed on opposite sides of the centre of a uniform metre rule, which is pivoted at its centre.

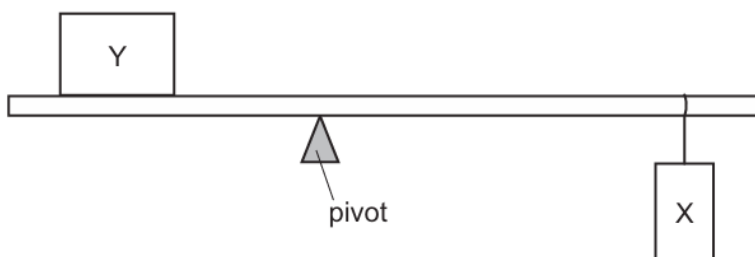


Which combination of masses and distances balances the rule?

	mass/g		distance/cm	
	X	Y	p	s
A	200	200	5	10
B	200	300	10	15
C	400	300	12	16
D	500	200	15	30

40. O/N/12/12/Q7

An object Y is in a fixed position on a rod. A weight X is moved and the position of a pivot is adjusted until the rod balances on the pivot, as shown.



The experiment is repeated in a region where the gravitational field strength is lower.

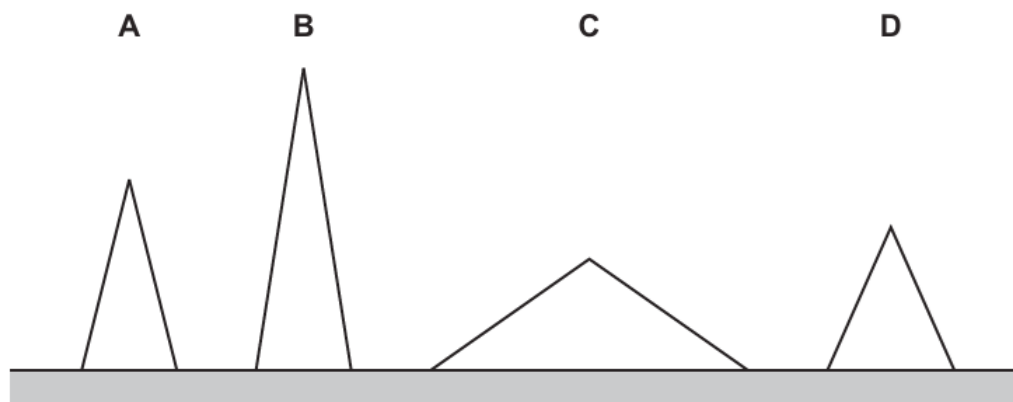
What is done to keep the rod balanced?

	pivot	X
A	move left	no movement
B	move right	move left
C	no movement	move right
D	no movement	no movement

41. O/N/12/12/Q8

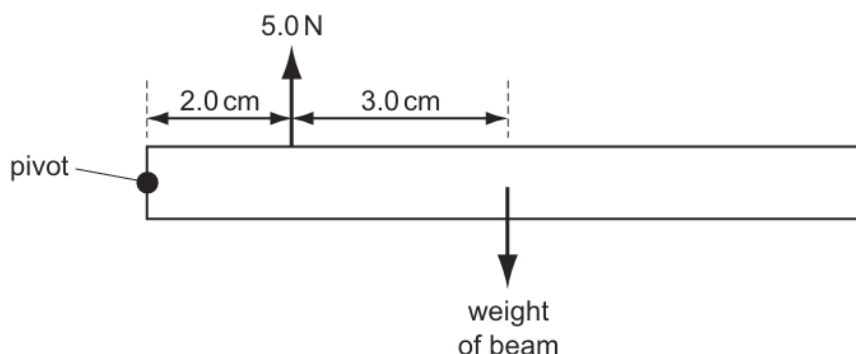
Four solid uniform cones have equal weight. They are placed on a bench as shown in the scale diagram.

Which cone is the most stable?



42. M/J/12/11/Q7

A beam pivoted at one end has a force of 5.0 N acting vertically upwards on it as shown. The beam is in equilibrium.

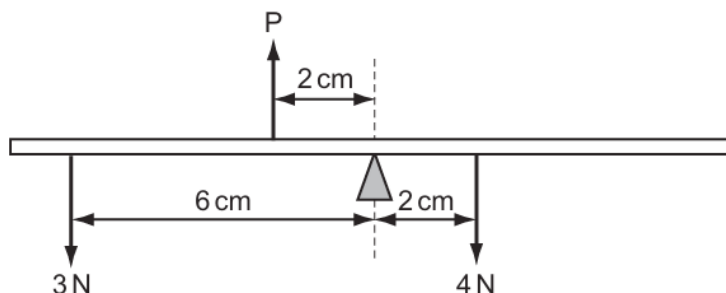


What is the weight of the beam?

- A** 2.0 N **B** 3.0 N **C** 3.3 N **D** 5.0 N

43. M/J/12/12/Q7

The diagram shows a uniform balanced beam, pivoted about its centre.



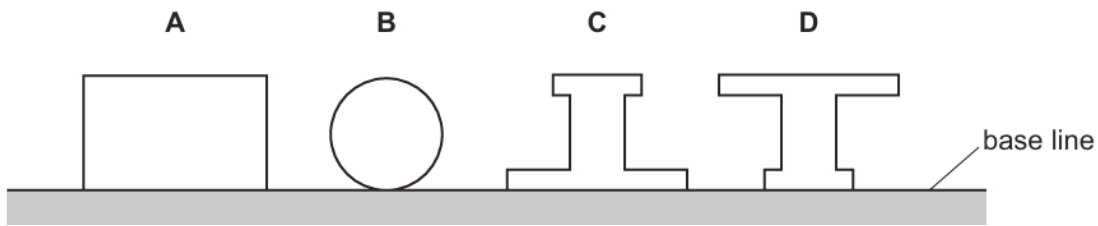
What is the value of force P?

- A** 5 N **B** 7 N **C** 10 N **D** 13 N

44. M/J/12/12/Q8

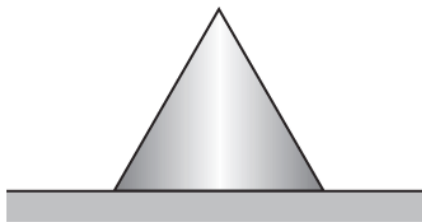
The diagram shows four shapes, cut from the same piece of card.

Which shape has its centre of mass nearest to the base line?



45. O/N/11/11/Q9

A metal cone with a circular base is placed on a flat surface.

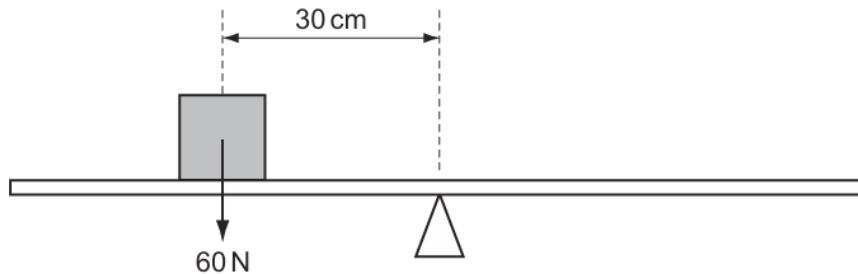


The stability of the cone depends on

- A its weight only.
- B the diameter of its base and the position of its centre of mass.
- C the diameter of its base only.
- D the position of its centre of mass only.

46. M/J/11/11/Q8

A uniform beam is balanced at its midpoint. An object is placed on the beam, as shown.



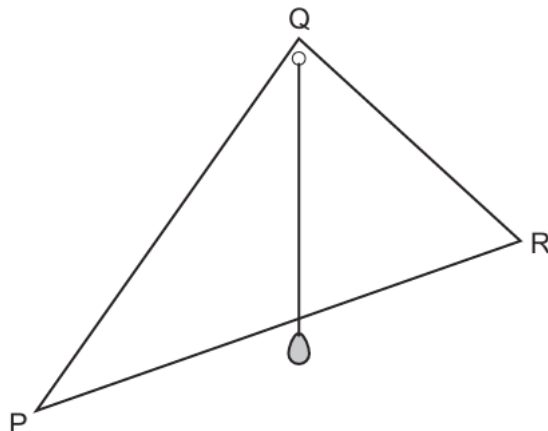
Which force will rebalance the beam?

- A 30 N acting upwards, 60 cm to the left of the midpoint
- B 30 N acting upwards, 60 cm to the right of the midpoint
- C 45 N acting downwards, 45 cm to the right of the midpoint
- D 90 N acting downwards, 20 cm to the left of the midpoint

47. M/J/11/11/Q9

A student finds the centre of mass of a triangular lamina PQR.

He drills a small hole at Q. He suspends the lamina from a pin through the hole at Q so that the lamina swings freely. He then hangs a plumb-line from the pin at Q, as shown. He marks the position of the plumb-line on the lamina.



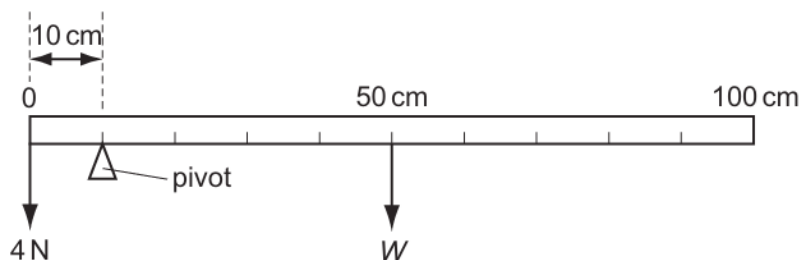
To determine the location of the centre of mass, the student then repeats the experiment but with one change.

What is the change?

- A** He suspends the lamina from the hole at Q, with R on the left and P on the right.
- B** He suspends the lamina from a pin through a hole at R.
- C** He uses a heavier weight on the plumb-line.
- D** He uses a longer plumb-line.

48. O/N/10/11/Q10

A uniform metre rule is balanced by a 4 N weight as shown in the diagram.

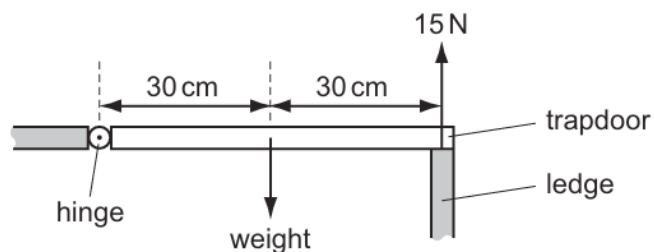


What is the weight W of the metre rule?

- A** 1 N
- B** 4 N
- C** 16 N
- D** 40 N

49. M/J/10/11/Q8

A wooden trapdoor is hinged along one side and, when closed, is supported on the other side by a ledge.



When the trapdoor is closed, the ledge exerts an upward force of 15 N on the trapdoor. The gravitational field strength is 10 N/kg.

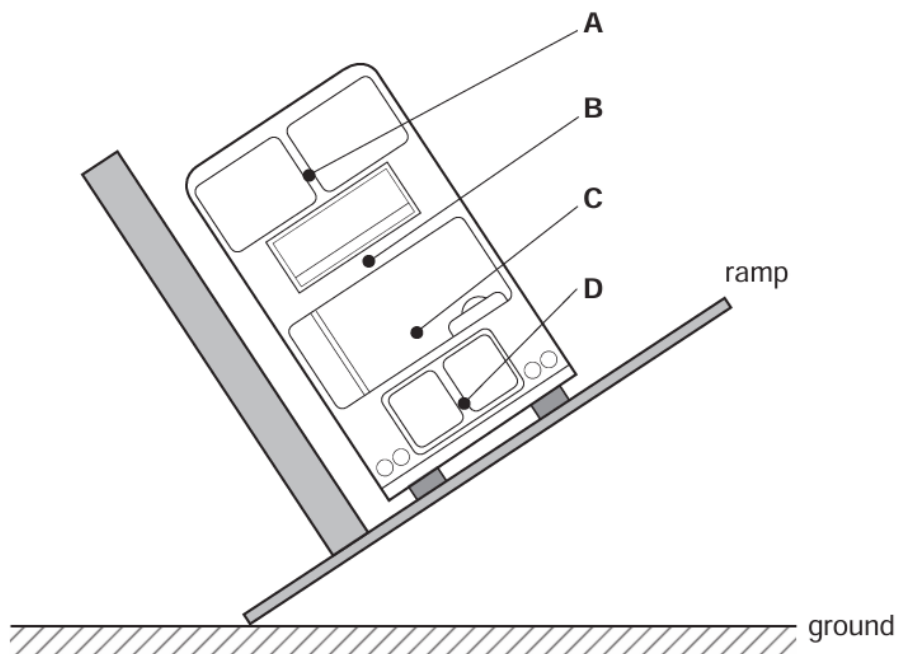
What is the mass of the trapdoor?

- A** 1.5 kg **B** 3.0 kg **C** 30 kg **D** 150 kg

50. O/N/09/Q8

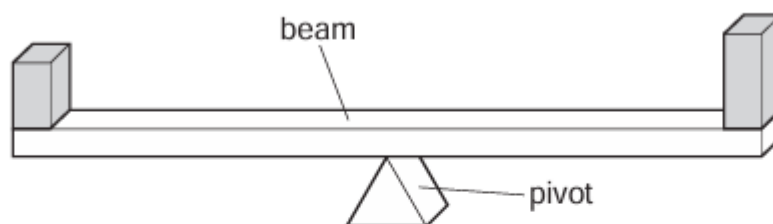
The stability of a bus is tested by tilting it on a ramp. The diagram shows a bus that is just about to topple over.

Where is the centre of mass of the bus?

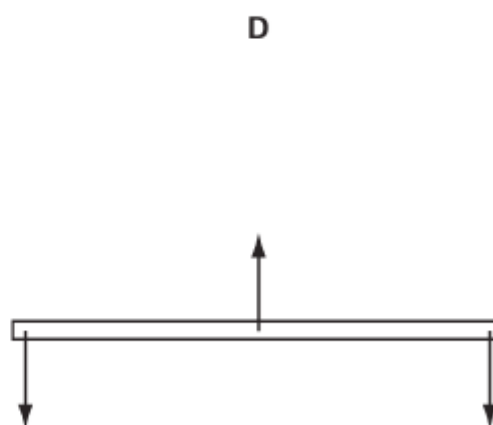
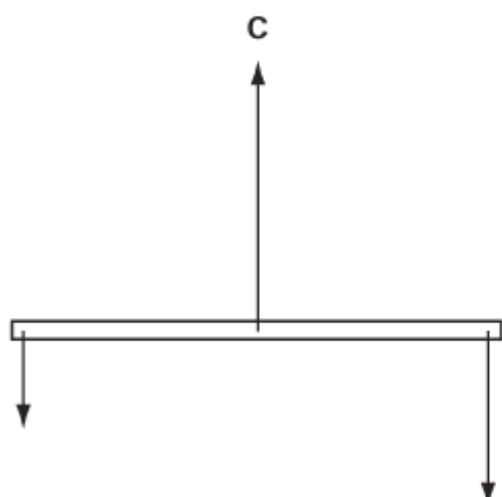
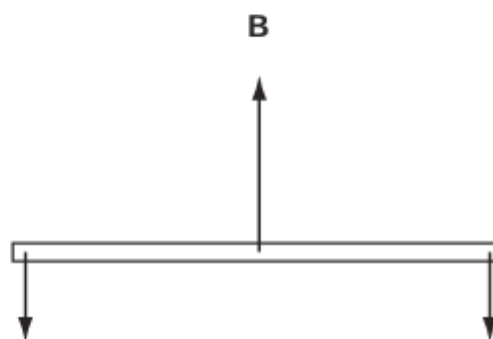
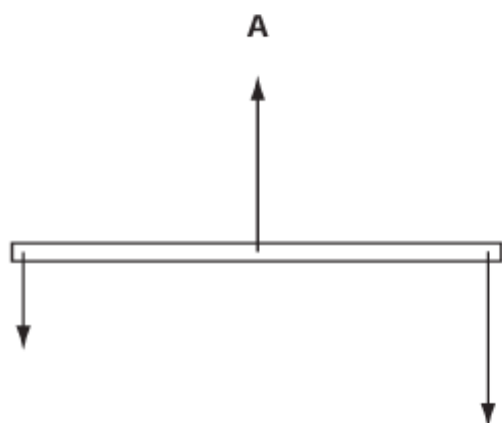


51. M/J/09/Q8

Two blocks are placed on a beam which balances on a pivot at its centre. The weight of the beam is negligible.

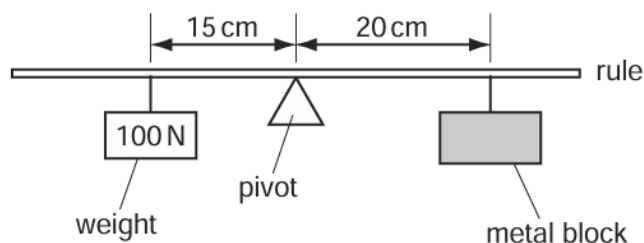


Which diagram shows the forces acting on the beam?
(The length of each arrow represents the size of a force.)



52. O/N/08/Q7

The diagram shows a uniform half-metre rule balanced at its mid-point.



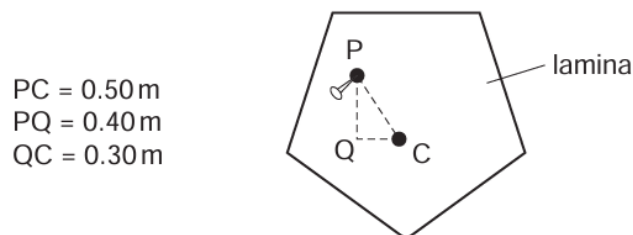
What is the weight of the metal block?

- A** 50 N **B** 75 N **C** 100 N **D** 150 N

53. M/J/08/Q8

A flat lamina is freely suspended from point P.

The weight of the lamina is 2.0 N and the centre of mass is at C.



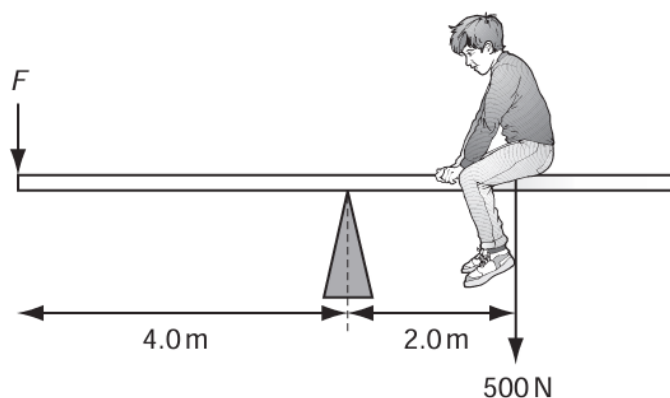
The lamina is displaced to the position shown.

What is the moment that will cause the lamina to swing?

- A** 0.60 N m clockwise
B 0.80 N m anticlockwise
C 1.0 N m clockwise
D 1.0 N m anticlockwise

54. O/N/07/Q7

The diagram shows a boy of weight 500 N sitting on a see-saw. He sits 2.0 m from the pivot.



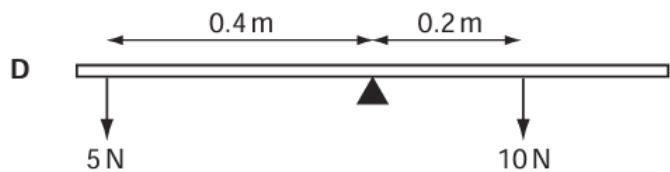
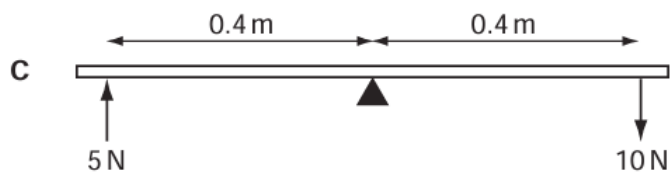
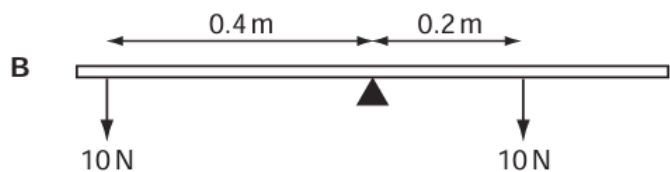
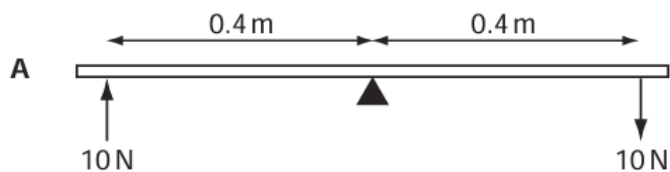
What is the force F needed to balance the see-saw?

- A** 250 N **B** 750 N **C** 1000 N **D** 3000 N

55. M/J/07/Q7

Forces are applied to a uniform beam pivoted at its centre.

Which beam is balanced?

**56. M/J/06/Q8**

If a nut and bolt are difficult to undo, it may be easier to turn the nut by using a longer spanner.

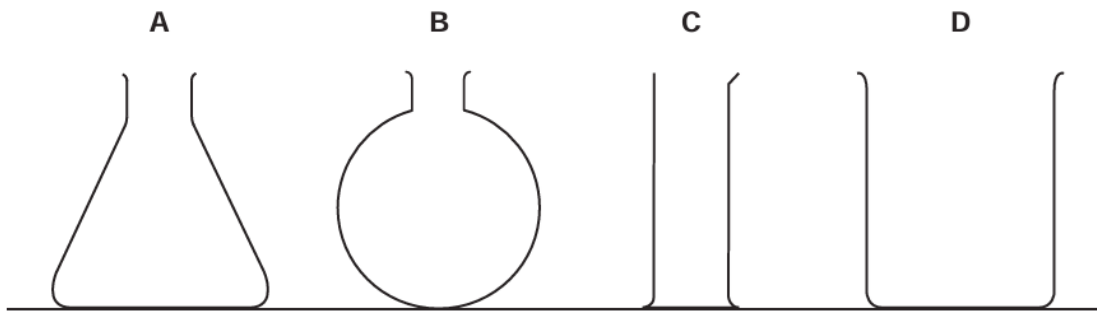
This is because the longer spanner gives

- A** a larger turning moment.
- B** a smaller turning moment.
- C** less friction.
- D** more friction.

57. M/J/06/Q9

Some containers are made from thin glass.

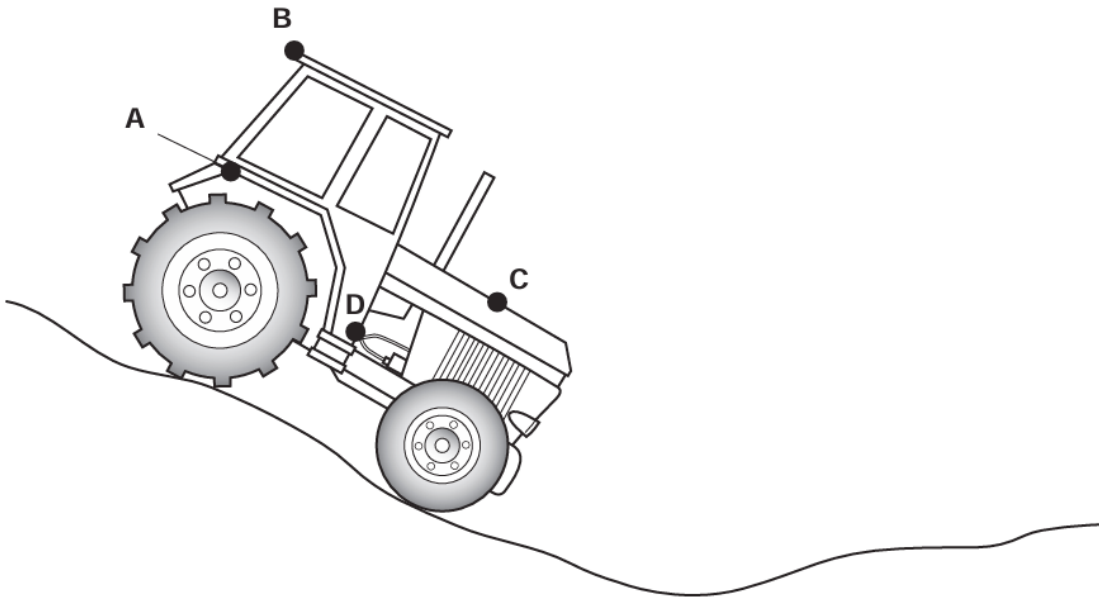
Which empty container is the **most** stable?



58. O/N/05/Q8

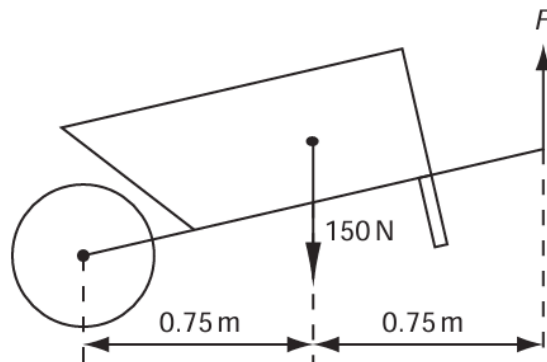
A tractor is being used on rough ground.

What is the safest position for its centre of mass?



59. M/J/05/Q6

The diagram shows a wheelbarrow and its load, which have a total weight of 150 N. This is supported by a vertical force F at the ends of the handles.



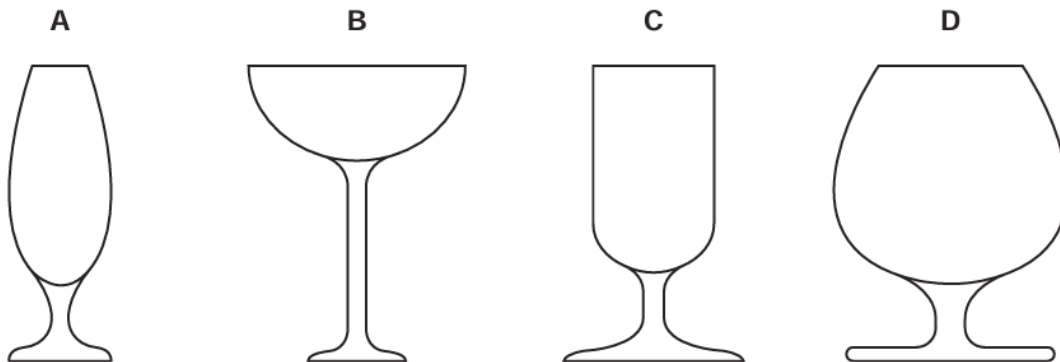
What is the value of F ?

- A** 75 N **B** 150 N **C** 225 N **D** 300 N

60. M/J/05/Q7

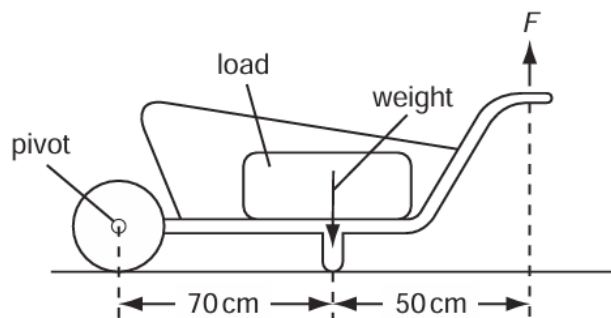
The diagrams show the cross-sections of different glasses.

Which one is the least stable when filled with a liquid?



61. O/N/04/Q8

A load is to be moved using a wheelbarrow. The total mass of the load and wheelbarrow is 60 kg. The gravitational field strength is 10 N/kg.

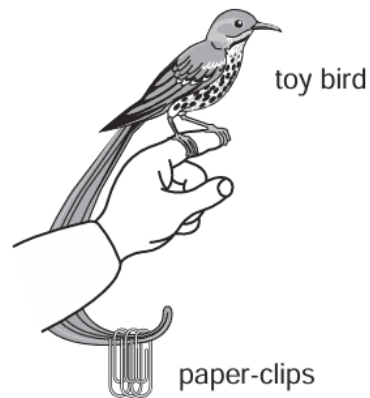


What is the size of force F needed just to lift the loaded wheelbarrow?

- A** 350 N **B** 430 N **C** 600 N **D** 840 N

62. O/N/04/Q9

A girl uses paper-clips to balance a toy bird on her finger as shown.



What is the effect of the paper-clips?

- A** They help to raise the centre of mass above her finger.
- B** They help to raise the centre of mass to her finger.
- C** They help to lower the centre of mass below her finger.
- D** They do not affect the centre of mass but increase the weight.